

AI ASSISTANT LAB-13.4

Rollno:2303A52140

Batch:41

Name: Ch.Rohith reddy

Task 1: Refactoring Data Transformation

Logic Scenario

Verbose loop used for numerical transformation.

Legacy Code

```
values = [2, 4, 6, 8, 10]
doubled = []
for v in values:
    doubled.append(v * 2)
print(doubled)
```

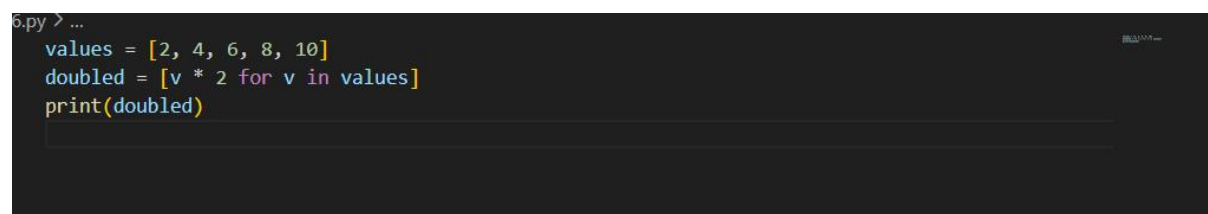
Output

[4, 8, 12, 16, 20]

Prompt used

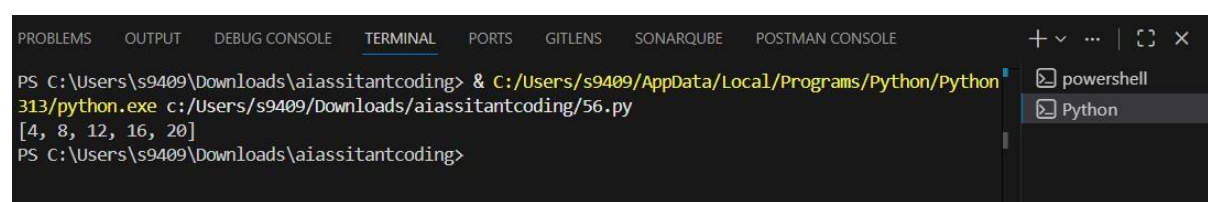
Refactor this Python loop into a more Pythonic approach using list comprehension while preserving output.

Refactored code



```
5.py > ...
values = [2, 4, 6, 8, 10]
doubled = [v * 2 for v in values]
print(doubled)
```

Output:



```
PS C:\Users\s9409\Downloads\aiassitantcoding> & C:/Users/s9409/AppData/Local/Programs/Python/Python313/python.exe c:/Users/s9409/Downloads/aiassitantcoding/56.py
[4, 8, 12, 16, 20]
PS C:\Users\s9409\Downloads\aiassitantcoding>
```

Justification

The legacy implementation used an explicit for loop and a mutable list to accumulate

results. While functionally correct, this approach reflects:

- Imperative programming style.
- Unnecessary boilerplate code.

- Reduced readability due to multiple lines for a simple transformation. The refactored version uses **list comprehension**, which is:

- More declarative.
- Expressive of intent (transform every element).
- Shorter and easier to read.
- Idiomatic Python

practice. From a performance standpoint:

- Time Complexity: $O(n)$ (unchanged).
- Space Complexity: $O(n)$ (unchanged).
- Slightly faster in C,Python due to internal optimization.

Thus, the refactoring improves clarity without altering computational complexity or output behavior.

Task 2: Improving Text Processing Code

Readability Scenario

Repeated string concatenation inside loop.

Legacy Code

```
words = ["Refactoring", "with", "AI", "improves", "quality"]
message = ""
for w in words:
    message += w + " "
print(message.strip())
```

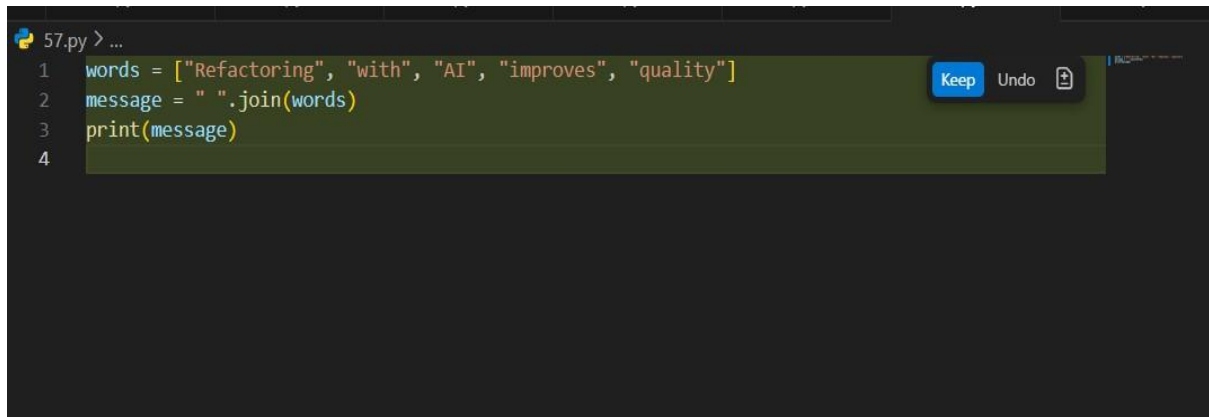
Output

Refactoring with AI improves

quality Ai prompt used

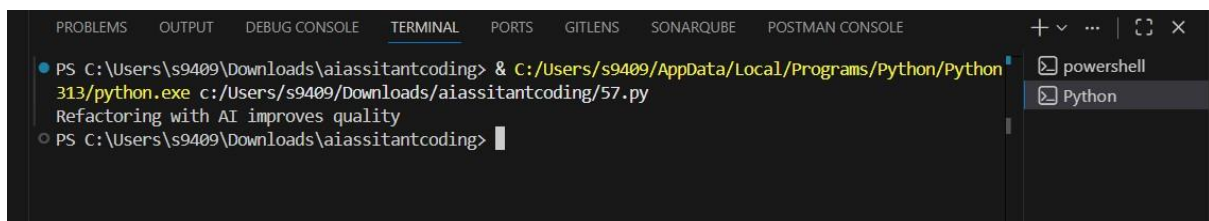
Suggest a more efficient method to build a sentence from a list of words in Python.

Refactored code



```
57.py > ...
1 words = ["Refactoring", "with", "AI", "improves", "quality"]
2 message = " ".join(words)
3 print(message)
4
```

Output:



```
PS C:\Users\s9409\Downloads\aiassitantcoding> & C:/Users/s9409/AppData/Local/Programs/Python/Python
313/python.exe c:/Users/s9409/Downloads/aiassitantcoding/57.py
Refactoring with AI improves quality
PS C:\Users\s9409\Downloads\aiassitantcoding>
```

Justification

The legacy approach performed string concatenation inside a loop:

```
message += w + " "
```

This is inefficient because:

- Strings in Python are immutable.
- Each concatenation creates a new string object.
- Time complexity can degrade toward $O(n^2)$ for large

lists. The refactored solution uses:

```
" ".join(words)
```

Advantages:

1. Optimized memory allocation.
2. Linear time complexity $O(n)$.
3. Cleaner and more readable expression of intent.
4. No need for `.strip()`.

This refactoring improves both **performance efficiency** and **code maintainability**.

Task 3: Safer Access to Configuration

Data Scenario

Manual key existence check in dictionary.

Legacy Code

```
config = {"host": "localhost", "port": 8080}
```

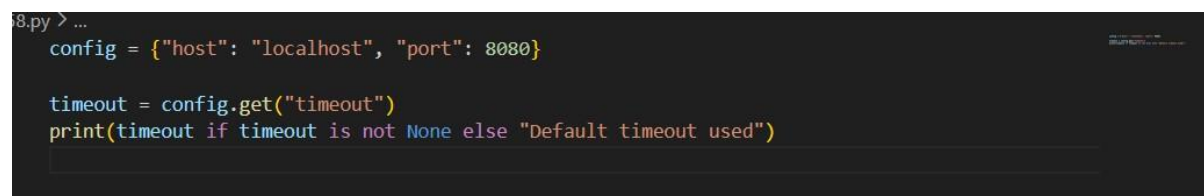
```
if "timeout" in config:
    print(config["timeout"])
else:
    print("Default timeout used")
```

Output

Default timeout used

AI Prompt Used

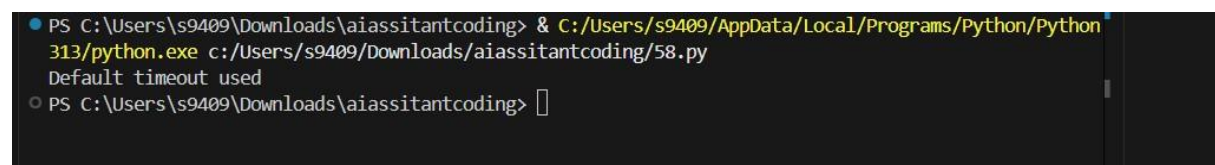
Refactor dictionary key checking using safer access methods in Python. Refactored code



```
58.py > ...
config = {"host": "localhost", "port": 8080}

timeout = config.get("timeout")
print(timeout if timeout is not None else "Default timeout used")
```

Output:



```
PS C:\Users\s9409\Downloads\aiassitantcoding> & C:/Users/s9409/AppData/Local/Programs/Python/Python313/python.exe c:/Users/s9409/Downloads/aiassitantcoding/58.py
Default timeout used
PS C:\Users\s9409\Downloads\aiassitantcoding>
```

Justification

The legacy code manually checked key presence: if "timeout" in config:

Issues:

- Verbose structure.
- Not scalable for multiple keys.
- Increases cognitive load.
- Risk of KeyError if incorrectly

accessed. The improved version uses:

```
config.get("timeout", "Default timeout used")
```

Advantages:

- Built-in safe accessor.
- Handles missing keys gracefully.
- Reduces conditional branching.
- More concise and expressive.

This improves robustness and follows defensive programming principles. Behavior remains identical:

- If key exists → return value.
- If key missing → return default message.

Task 4: Refactoring Conditional Logic for Scalability

Scenario

Multiple `if-elif` statements for operation handling.

Legacy Code

```
• action = "divide"
  x, y = 10, 2

  if action == "add":
      result = x + y
  elif action == "subtract":
      result = x - y
  elif action == "multiply":
      result = x * y
  elif action == "divide":
      result = x / y
  else:
      result = None

  print(result)
```

Output

```
5.0
```

Ai prompt used

Replace multiple `if-elif` conditions with a more scalable approach using mapping techniques.

Refactored code

```
59.py > ...
1 action = "divide"
2 x, y = 10, 2
3
4 operations = {
5     "add": lambda a, b: a + b,
6     "subtract": lambda a, b: a - b,
7     "multiply": lambda a, b: a * b,
8     "divide": lambda a, b: a / b,
9 }
10
11 operation = operations.get(action)
12 result = operation(x, y) if operation is not None else None
13
14 print(result)
15
```

Output:

```
PS C:\Users\s9409\Downloads\aiassitantcoding> & C:/Users/s9409/AppData/Local/Programs/Python/Python313/python.exe c:/Users/s9409/Downloads/aiassitantcoding/59.py
5.0
PS C:\Users\s9409\Downloads\aiassitantcoding> 
```

Justification

The legacy implementation uses a chained if-elif-else structure. Problems include:

- Poor scalability.
- Hard to extend.
- Violates Open-Closed Principle.
- Code duplication risk when logic grows.

The refactored version introduces a **dispatch dictionary (operation mapping)**:

```
operations = {
    "add": lambda a, b: a + b,
    ...
}
```

Benefits:

1. Constant-time lookup $O(1)$.
2. Extensible design (add new operation easily).
3. Cleaner separation of logic.
4. Reduced cyclomatic complexity.
5. Follows functional programming style.

This design pattern improves maintainability and scalability in larger systems.

Task 5: Simplifying Search Logic in Collections

Scenario

Manual loop for searching item in list.

Legacy Code

```
inventory = ["pen", "notebook", "eraser",  
"marker"] found = False
```

```
for item in inventory:
```

```
    if item ==  
    "eraser":  
        found =  
        True break
```

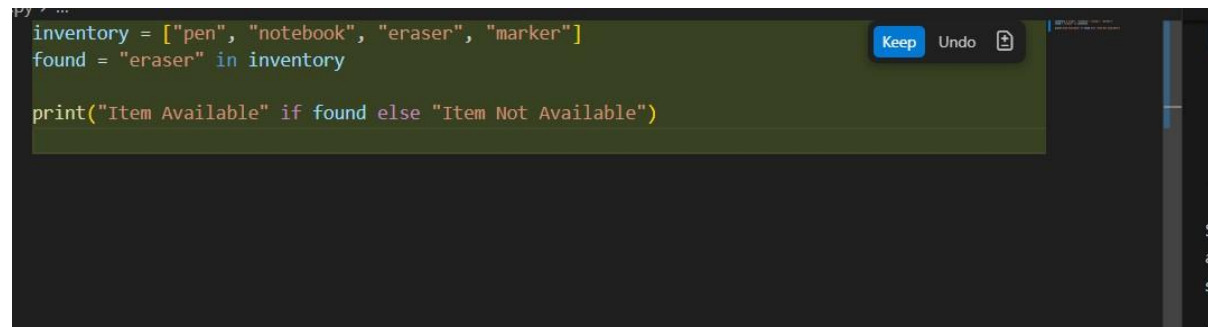
```
print("Item Available" if found else "Item Not Available")
```

Output

Item Available

Ai prompt used

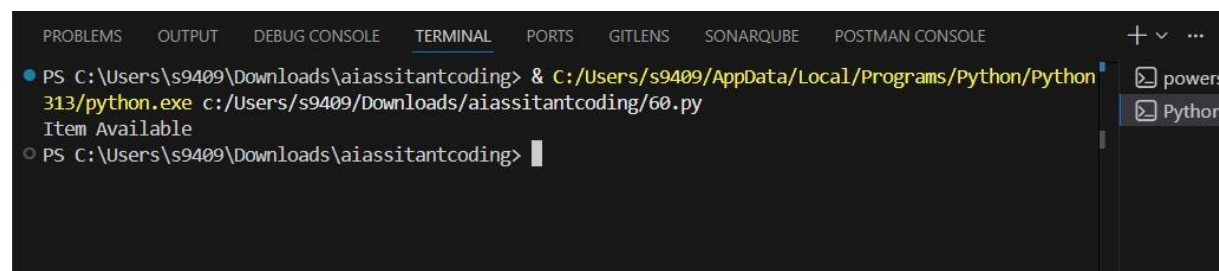
Refactor this manual search loop into a more concise Pythonic approach. Refactored code

A screenshot of a code editor with a dark theme. The code is as follows:

```
inventory = ["pen", "notebook", "eraser", "marker"]  
found = "eraser" in inventory  
  
print("Item Available" if found else "Item Not Available")
```

At the top right of the code editor, there are buttons for "Keep", "Undo", and a plus icon. The code is highlighted in a light green color.

Output:

A screenshot of a terminal window with a dark theme. The terminal shows the following commands and output:

```
PS C:\Users\s9409\Downloads\aiassitantcoding> & C:/Users/s9409/AppData/Local/Programs/Python/Python313/python.exe c:/Users/s9409/Downloads/aiassitantcoding/60.py  
Item Available  
PS C:\Users\s9409\Downloads\aiassitantcoding>
```

The terminal window has tabs for "PROBLEMS", "OUTPUT", "DEBUG CONSOLE", "TERMINAL", "PORTS", "GITLENS", "SONARQUBE", and "POSTMAN CONSOLE". The "TERMINAL" tab is active.

Justification

The legacy implementation manually iterates:

for item in inventory:

Problems:

- Redundant boolean flag.
- Unnecessary loop control.
- More lines than required.
- Less expressive

intent. Refactored version:

"eraser" in inventory

Advantages:

1. Direct membership testing.
2. Cleaner and more semantic.
3. Built-in optimization.
4. Maintains same $O(n)$ complexity.
5. Eliminates control flow clutter.

This enhances readability and adheres to Pythonic principles.